Following the Compute Money: An On-Chain Forensic Analysis of GPU-Backed Credit and Decentralized Compute Markets

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Code and frozen data: https://github.com/uwecerron/covalent_onchain_compute

***Abstract*—Several synthetic-dollar protocols take stablecoin deposits and lend them to companies that operate GPU clusters. The hardware, borrowers' revenues and loan agreements are off-chain, while loan and vault transactions are recorded on a public chain. We reconstruct USD.AI on Arbitrum from 55,954 decoded events between 2025-05-12 and 2026-10-22, using the Covalent GoldRush API and signatures from protocol source code. We identify 39 loans to 13 borrower addresses, with $290.1 M originated and $182.2 M outstanding at the study cutoff. No breach or liquidation event was recorded; most active loans have not reached maturity. Protocol-hosted metadata describes about 4,560 pledged GPUs at 66.8% stated loan-to-value, but does not independently establish custody or realizable collateral value. Annualized deposit-side NAV growth was 8.17%. An event-based cash-to-accrued-yield ratio was 93.7%, but borrower interest and fees contributed only 28.4% of credited yield; two exceptional transfers materially raise the broader ratio. Completed redemption requests had a median wait of 13.0 days; pending requests remain censored. An exact-block accounting check put the reconstructed loan book within 0.22% of the vault's reported balance, with the residual not yet itemized. At that snapshot, 23.2% of reported assets was attributed to two pre-funded escrows without matching on-chain originations; transfers to external wallets do not establish the legal status or recoverability of the claims. We also measure $4.47 M in payments on Akash and Livepeer. Comparing those payments with $290.1 M of loan originations is descriptive, not a like-for-like measure of compute demand. The accompanying data document the reconstruction; operational cash flows and legal collateral rights require separate verification.**

***Index Terms*—DeFi, stablecoins, GPU finance, on-chain forensics, real-world assets, decentralized compute, ERC-4626, ERC-7540**

I. Introduction

Several synthetic-dollar protocols now take stablecoin deposits on-chain and lend them to companies that operate GPU clusters, paying depositors from the interest. USD.AI is one of the larger ones. It has a $100 M credit facility from Bullish [1], and its tokens have been proposed as collateral on Aave [2]. Futures on GPU rental prices are also being built so lenders can hedge the same risk [3]. Lenders, risk curators and auditors are pricing GPU-backed credit, but nobody has published an independent account of what the chain shows about it.

The GPUs, the borrowers' revenues and the loan agreements are off-chain. The recorded liabilities are not. Deposits, loan disbursements, repayments, fees, redemption requests and bridge transfers leave event logs on a public chain. Those logs cannot prove that a GPU exists. They can reconstruct recorded loan terms and payments, yield transfers, token custody and completed redemption times. They cannot establish beneficial ownership, collateral perfection or ultimate borrower cash flow. A depositor or a risk curator needs both kinds of evidence before relying on the off-chain story.

We rebuild USD.AI from raw logs using the Covalent GoldRush API [4] and the protocol's source code [5], and ask seven questions:

- RQ1 Where does deposited money go?
- RQ2 Where does the yield come from, and is NAV growth backed by cash from borrowers?
- RQ3 Who holds the yield-bearing token?
- RQ4 How concentrated are the borrowers?
- RQ5 How do redemptions behave, and do they follow GPU rental prices?
- RQ6 Which chains does the protocol live on?
- RQ7 How do loan originations compare descriptively with payments on Akash and Livepeer, two distinct on-chain compute venues?

We also check every headline number against the vault's own accounting at a single block, and we trace one kind of vault outflow, escrow pre-funding, to where the money ended up.

Contributions.

1) An open, tested pipeline that decodes every event from source-code signatures, follows loans through a V1 to V2 migration and through refinancings, and reproduces all results offline from a hash-verified dataset.
2) A reconstruction of a GPU-backed loan book from chain data: 39 loans to 13 borrowers, $290.1 M originated and $182.2 M outstanding.
3) A cash-coverage test that separates yield received in cash from yield credited through NAV.
4) A reconstruction of the collateral from the protocol's hardware-token metadata: model, unit count, stated value and loan-to-value for every active loan.
5) Measurements of holder concentration, the redemption queue and supply across five chains.
6) A reconciliation showing that 23.2% of reported vault assets is an escrow claim on funds that were redeemed to PYUSD and sent onward before any matching loan existed.

7) A descriptive comparison of GPU credit originations with spending on two decentralized compute venues over the same period; these are different economic quantities, not an estimate of total compute demand.
8) Monitoring indicators that lenders and curators can run with the same code.

Earlier empirical work on DeFi lending studied interest rates and liquidations in over-collateralized markets with on-chain collateral [6]–[8]. Stablecoin research models peg stability and run risk [9]–[11]. A GPU-backed dollar fits neither. Its collateral is off-chain and illiquid, its assets are term loans, and holders exit through a queue that an administrator services. Our earlier paper modeled the economic life of GPUs as assets [12]. This one looks at the debt financed against them.

II. Protocol Background

USD.AI issues two tokens on Arbitrum, its hub chain [13]. *USDai* is minted one-for-one against approved stablecoins (PYUSD, M^0's wM and USDC over our window) by KYC'd institutions through `deposit()` and burned back to a stablecoin through `withdraw()`. It does not accrue yield. *sUSDai* is a vault over USDai. Deposits are synchronous under ERC-4626 [14]. Exits are asynchronous under ERC-7540 [15]: a holder calls `requestRedeem()`, which burns the shares and joins a first-in, first-out queue. An administrator role then services the queue at fixed windows (the documentation gives 30 days as an example) by calling `serviceRedemptions()`. The vault prices deposits at an *optimistic* NAV and redemptions at a *conservative* NAV, and the two coincide only when no loans are active [13].

Vault capital reaches borrowers through a lending stack (Fig. 1). The vault commits USDai to a deposit or escrow timelock keyed by a `loanTermsHash`. A *Loan Router* originates the loan, mints lender position NFTs, and routes monthly repayments of principal and interest back to tranche lenders. Collateral is a wrapped bundle of GPU-backed tokens, and loans are non-recourse to the borrower's parent except for "bad acts" [13]. A second yield source, a *Base Yield Escrow*, holds PYUSD issuer incentives that are periodically harvested into the vault as newly minted USDai. Earlier in our window, yield on M^0's wM reserve was also minted into the vault as base yield. Both tokens are bridged with LayerZero burn-and-mint OFTs [16] to Ethereum, Base, Plasma and Solana. Every core contract can be paused by an administrator role, and operations are submitted by a multisig [13].

The source code and its full git history are public [5]. We use them in two places. First, every event signature we decode comes from that code, including signatures that were later renamed or removed. Second, the history shows a V1 to V2 Loan Router upgrade (commit `43dd6d5`, 2026-05-29) and a temporary `migrateLoan()` path, which we see executed on-chain between 2026-06-26 and 2026-07-14.

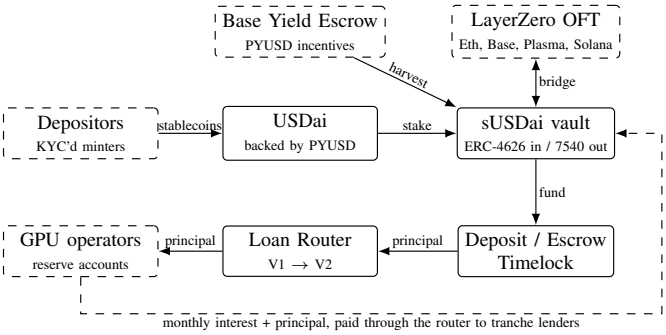


Fig. 1. USD.AI as reconstructed in this study. Solid arrows are flows we measure directly from Arbitrum event logs. Dashed boxes are parties or venues whose internal state is not on the hub chain.

III. Data and Methodology

A. Data acquisition

All chain data comes from the Covalent GoldRush REST API [4] (Table I). The study window runs from the vault's deployment block (Arbitrum block 335,730,936, 2025-05-12) to block 517,316,432 (2026-10-22). The first vault deposit, and so the first NAV observation, is on 2025-05-14. Livepeer and Akash are measured over the same dates. We pull three kinds of data:

- **Complete transaction histories**, with decoded logs, for the nine lending-stack contracts (Table I).
- **Event scans** of the USDai and sUSDai contracts for 15 event types. The topic endpoint accepts at most 10^6 blocks per call, so the window is split into 182 non-overlapping block windows. Windows that are fully in the past are cached; the open window is always refetched.
- **Snapshots**: current holders of USDai and sUSDai on four chains, and daily USD prices.

We also use the weekly median H100 on-demand rental price from a frozen GetDeploying snapshot (CC BY 4.0) [17], the same file used in [12]. Its SHA-256 matches the original.

B. Decoding

We decode raw logs ourselves rather than rely on third-party ABI coverage. For each event we compute $topic_0 = \mathrm{keccak256}(signature)$ from the published source. A log is accepted only if its topic count matches the number of indexed parameters, which, for example, rejects ERC-721 `Transfer` events. The data field is then ABI-decoded. V2 originations, migrations and refinancings carry the full `LoanTermsV2` struct: borrower, currency, collateral token IDs, tranche lenders, amounts and per-second rates, repayment schedule and fee specifications. We decode it to recover terms that V1 only kept in calldata. The decoder recognizes 55,954 events. Two topic hashes on the V1 router match no signature in the repository history; we count them and exclude them.

C. Stitching the loan book

A loan changes identity when it is migrated or refinanced, because each event retires one `loanTermsHash`

TABLE I
DATA SOURCES AND WHAT EACH ONE SUPPLIES

Source	Endpoint or method	What we pull
GoldRush	`transactions_v3`	Full decoded history of the 9 lending contracts: Loan Router V1 and V2, deposit, escrow and collateral timelocks, reserve-account factory, collateral wrapper, base yield escrow
GoldRush	`events/topics`	15 vault and USDai event types over 182 block windows; Livepeer `TicketBroker` ticket payments
GoldRush	`token_holders_v2`	Current holders of USDai and sUSDai on 4 chains; monthly sUSDai holders on Ethereum at past blocks
GoldRush	`historical_by_addresses_v2`	Daily market prices of sUSDai, USDai, PYUSD and WETH
GoldRush	`transactions_summary`	Activity of top holders and borrowers
GoldRush	`address/{a}/activity`	Every chain each borrower and top holder has used
GoldRush	`balances_v2`	Current holdings of traced wallets
Arbitrum node	`eth_call`	Vault accounting views at one block
Arbitrum node	`eth_getLogs`	Independent re-pull of every event set, to check GoldRush (Section IV-F)
Arbitrum node	`tokenURI`, `ownerOf`	Hardware-token metadata links and holders for every pledged token
USD.AI metadata host	HTTPS	The protocol's own record for each hardware token: model, units, stated value, useful life
Solana node	`getTokenSupply`	sUSDai and USDai supply on Solana
Akash Console API	REST	Akash spend, leased GPUs, providers, GPU prices
GetDeploying	frozen file	Weekly median H100 rental price (CC BY 4.0)

and creates another. We build a forest in which each new hash points to its predecessor and resolve every repayment to the root origination. Migrated loans name a per-borrower *reserve account* as the new borrower. The factory event `ReserveAccountCreated(account, borrower)` maps each reserve account back to the underlying borrower. Each hash keeps the currency of its own terms: V1 loans were denominated in PYUSD or USDC (6 decimals) and migrated loans in USDai (18 decimals). Outstanding principal is the principal of the current terms minus principal repaid under those terms. The tranche rate r is per-second and scaled by 10^{18}, so the annual rate is $\mathrm{APR} = r \cdot 31{,}536{,}000/10^{18}$. Loans below $1,000 are test loans and are excluded.

D. Vault accounting

Let $\mathcal{D}_d$ be the set of sUSDai `Deposit` events on day d. We define the deposit-side NAV and the global share supply as

$$\mathrm{NAV}_d = \mathrm{median}_{e \in \mathcal{D}_d} \frac{assets_e}{shares_e}, \tag{1}$$

$$S_d = \sum_{e \le d} shares_e^{\text{mint}} - \sum_{e \le d} shares_e^{\text{request}}. \tag{2}$$

NAV is forward-filled on days without deposits. Shares are burned when a redemption is requested, not when it is paid. Bridging moves shares between chains without changing S_d; we validate S_d against the sum of the four holder snapshots. The yield accrued to holders in a period P is

$$Y_P = \sum_{d \in P} S_{d-1} \left(\mathrm{NAV}_d - \mathrm{NAV}_{d-1}\right), \tag{3}$$

and the cash that reached the vault is $C_P = I_P + B_P + F_P + E_P$. Here I_P is loan interest paid to the vault (`LenderRepaid` with the vault as lender), B_P is base yield harvested into the vault, F_P is fees paid to the vault, and E_P is amounts the contracts booked as escrow interest on withdrawal or cancellation, net of the 10% router admin fee. We separately identify recurring sources and unusually large one-off transfers rather than infer their economic purpose from their event names. The *cash coverage ratio* $\kappa = C/Y$ asks how much of the NAV growth holders were credited with arrived as cash. A ratio below one means part of the reported yield is accrued, not received. That is normal for loans that pay monthly, but it should close over time.

E. Holders, queue and GPU prices

For each chain we report holder count, top-k shares, the Herfindahl-Hirschman index (HHI) and the Gini coefficient. Top holders are labeled using the contract metadata names GoldRush returns for logs they emit. Redemption requests are matched to `RedemptionProcessed` events by request ID to obtain wait times and pending shares. Finally, we compute Spearman's ρ between weekly requested shares and the H100 rental price, both its level and its week-over-week change. The sample is small, so this is descriptive and not causal.

F. Reconciliation with on-chain state

Event-derived numbers can be wrong in ways that are hard to see from the events alone. We therefore read the vault's own accounting views (`totalAssets`, `totalShares`, `bridgedSupply`, `depositTimelockBalance`, `loanRouterBalances`, the deposit and redemption share prices, and related views) with `eth_call` at a single Arbitrum block, 517,316,432. GoldRush does not expose arbitrary contract calls, so this one step uses the public Arbitrum RPC. Every reconciliation in Section IV-E compares an independently computed event total to one of these views.

G. Following escrowed funds

The vault can commit USDai to an escrow timelock against a `loanTermsHash` before any loan with that hash exists. When an escrow deposit has no matching origination, we follow the stablecoins. Starting from the escrow facility that received the USDai, we pull complete transaction histories with GoldRush and decode every ERC-20 `Transfer` of USDai, PYUSD and USDC, plus USDai `Withdrawn` events, which mark redemption of USDai into PYUSD. We expand two hops and stop at wallets with more than 1,500 transactions, which we treat as services. We label wallets only by their on-chain role. We do not attribute them to off-chain entities.

H. Decentralized compute markets

To put GPU credit next to realized on-chain demand for compute, we measure two live markets. *Livepeer* pays for video transcoding and AI inference with probabilistic micropayments on Arbitrum. A gateway funds a deposit in the `TicketBroker` contract and an orchestrator redeems winning tickets. Each `WinningTicketTransfer(sender, recipient, amount)` event is ETH that actually moved from a payer to a provider. We scan it over the same window and price it at the daily WETH price from GoldRush. *Akash* runs on its own Cosmos chain, which GoldRush does not index. We take its daily USD spend, active GPU count, provider count and per-model GPU prices from the public Akash Console API, and label them as such. Finally, we compare GPU rental prices on Akash with the centralized-cloud H100 median in the GetDeploying snapshot.

I. Where each number comes from

Event logs, transaction histories, holder lists, prices, cross-chain activity and wallet balances come from GoldRush. The vault's accounting views at one block come from `eth_call` on a public Arbitrum node, Solana token supply from a public Solana node, Akash figures from the Akash Console API, and GPU rental prices from GetDeploying. Section IV-F re-checks the GoldRush event data against a public node.

J. Reproducibility

Raw responses are canonicalized into `data/frozen` with a SHA-256 manifest. `reproduce.py` regenerates every number, table and figure in this paper offline; numbers enter the text through generated LaTeX macros. The repository has 41 unit tests, including tests that the API key cannot leak, and runs them in CI.

TABLE II
USDAI MINTING AND REDEMPTION BY BACKING ASSET

Backing asset	Minted	Redeemed
PYUSD	\$757.80 M	\$1,140.29 M
wM (M^0)	\$542.73 M	\$33.95 M
USDC	\$353.25 M	\$225.99 M
USDT	\$0.04 M	\$0.00 M
Total	\$1,653.8 M	\$1,400.2 M

IV. Results

Fig. 2 lists the main measured quantities. We take the research questions in order.

A. RQ1: Where the money goes

Over the window, institutions minted \$1,653.8 M of USDai and burned \$1,400.2 M. Minting was backed by PYUSD, by M^0's wrapped \$M (wM), by USDC and, marginally, by USDT (Table II). Not every burn is an exit. Loan disbursements also burn USDai into the borrower's currency at origination, and the escrow redemption in Section IV-G is a burn too. Of the USDai still outstanding, most sits in the vault as idle deposit balance rather than in loans (Section IV-E).

From that pool the Loan Routers originated 39 loans (excluding 2 test loans below \$1,000) for \$290.1 M of principal to 13 distinct borrower addresses. 17 loans have a recorded migration event and four were refinanced into new terms; root originations are counted once. 18 loans have been repaid in full. None has a recorded breach or liquidation event in the observed period; that is not a lifetime default estimate. At 2026-10-22, \$182.2 M was outstanding. Borrowers have paid \$6.94 M of interest, \$108.1 M of principal and \$515 k of repayment fees. Fig. 3 shows originations by month and borrower.

B. What the collateral is

Every loan's collateral is an ERC-721 token from the protocol's hardware contract ("USD.AI Tokenized Hardware"), either directly or wrapped in a bundle. Each minted token's `tokenURI`, read from the contract, points to a metadata record the protocol publishes with the server model, the number of units, a stated collateral value, a useful life and a depreciation start. We read token IDs from decoded `LoanTermsV2` and bundle transfer logs, and URIs and owners by `eth_call`. Including two additional records served by the protocol's metadata host for token IDs not minted at retrieval, the host returned 87 records (retrieved 2026-10-27). The records are the protocol's own statements, hosted off-chain and changeable, so what follows is what USD.AI says the collateral is, not an appraisal. Unminted records are excluded from active pledged collateral.

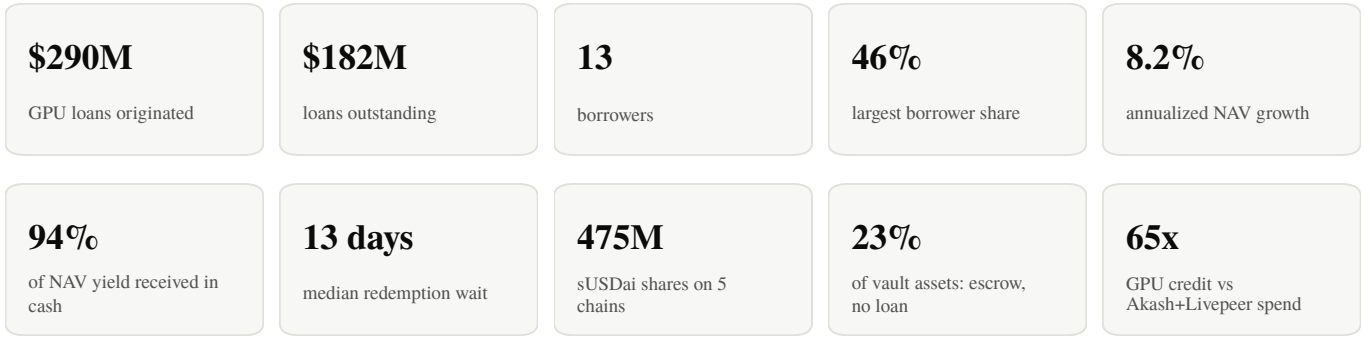


Fig. 2. Key quantities at a glance over 2025-05-12 to 2026-10-22. Cash coverage includes the exceptional transfers separated in Table IV; collateral value is protocol-stated.

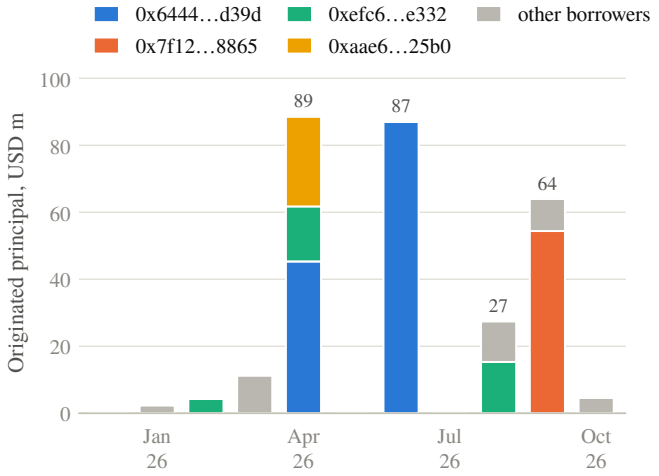


Fig. 3. Monthly originated principal by borrower. The four largest borrowers are shown separately.

TABLE III
COLLATERAL BEHIND ACTIVE LOANS, AS REPORTED IN PROTOCOL-HOSTED HARDWARE METADATA. GPU COUNTS USE SERVER-MODEL ASSUMPTIONS WITH UNRESOLVED MIXED-UNIT ENTRIES; WORKSTATION UNITS ARE NOT INCLUDED IN THE AGGREGATE GPU COUNT.

Borrower	Hardware (units)	GPUs	Stated value	Outstanding	LTV
`0x7f12...8865`	B300 (130)	1,040	$77.7 M	$54.4 M	70%
`0x6444...d39d`	B200 (96)	768	$42.7 M	$31.2 M	73%
`0xaae6...25b0`	B300 (72)	576	$38.3 M	$22.4 M	58%
`0xefc6...e332`	B300 (34)	272	$19.2 M	$14.9 M	78%
`0xefc6...e332`	H200 (75)	600	$20.5 M	$14.1 M	69%
`0x6444...d39d`	B200 (32)	256	$14.1 M	$10.5 M	74%
`0x9243...1960`	B300 (24)	192	$12.0 M	$9.6 M	80%
`0xd08f...359a`	B200 (32)	256	$14.2 M	$7.3 M	51%
`0xc991...5398`	B200 (38)	304	$15.8 M	$7.1 M	45%
`0xc991...5398`	B200 (16)	128	$7.1 M	$4.6 M	65%
`0xefc6...e332`	B300/RTX 6000 (27)	72	$6.1 M	$3.4 M	56%
`0x8db6...2fb9`	B200 (36)	64	$3.3 M	$1.8 M	54%
`0x6726...be31`	H200 (8)	32	$0.9 M	$0.4 M	43%
8 smaller loans	mixed (10)	n/a	$0.97 M	$0.59 M	60%
All active	(630)	4,560	$272.8 M	$182.2 M	67%

Table III and Fig. 4 list the active loans. The metadata describes NVIDIA B300, B200 and H200 servers and, in the small early loans, RTX 6000 and RTX 5090 workstations. Our aggregate count is 630 hardware units and approximately 4,560 GPUs, with $272.8 M of stated value against $182.2 M outstanding: an indicated book loan-to-value of 66.8%, ranging from 43% to 80% by loan. The eight-GPU-per-server assumption does not explain every mixed or aggregated row in Table III; those records need a model-by-model reconciliation before treating the aggregate as an independently verified fleet count. Every record states a useful life of 1080 and 1095 days, about three years, with depreciation starting "on borrow". Under straight-line depreciation and no principal reduction, an 80% initial LTV would reach 100% after roughly 7.2 months. The date changes with amortization, market resale values and any contractual credit protection, which are not established by the metadata [12].

One token stands out. NVIDIA B300 (288), 288 servers with a stated value of $140.2 M, secured the largest borrower's $87.1 M loan until that loan was repaid in full on 2026-10-01, eight days after the $128.9 M escrow deposits. The token now sits unpledged in the borrower's reserve account. The host also serves records for two unminted GB200 NVL72 token IDs, 1012 and 1013, with $161.1 M in combined stated value. A refinancing in progress is one possible explanation for this sequence, but no matching new loan or legal commitment can be verified from these observations. Neither unminted record proves the existence of pledged collateral.

C. Maturities and payment status

The 21 active loans have original terms of 2.1 to 3.0 years. Weighted by outstanding principal, the book has 2.6 years left to run: $13.3 M matures in 2028 and $168.9 M in 2029. Against that, completed redemptions have a 13-day median wait (Section IV-J). The longest gap since a loan's last recorded repayment is 25 days, inside the reported monthly schedule, and 0 loans are flagged late by the event-based test. The oldest active loan is under a year old. The absence of observed delinquency is therefore evidence about a young, right-censored loan book, not its full-cycle credit performance.

D. RQ2: Where the yield comes from

The deposit-side NAV of sUSDai rose from 1.0003 to 1.1202 (Fig. 5). That is 8.17% annualized over the full window

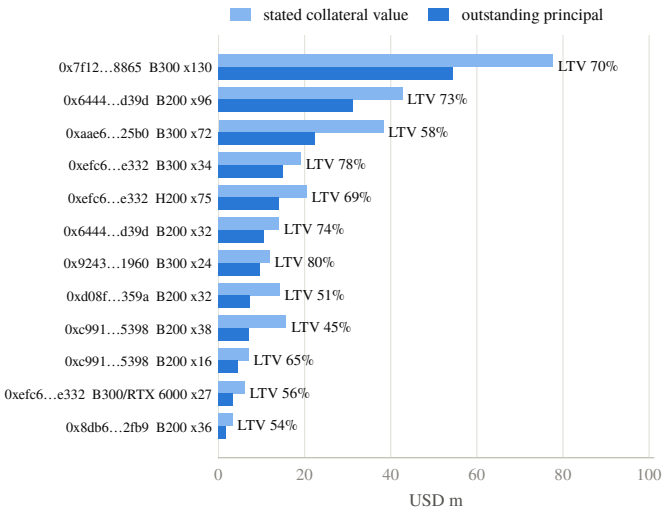


Fig. 4. Stated collateral value against outstanding principal for the largest active loans.

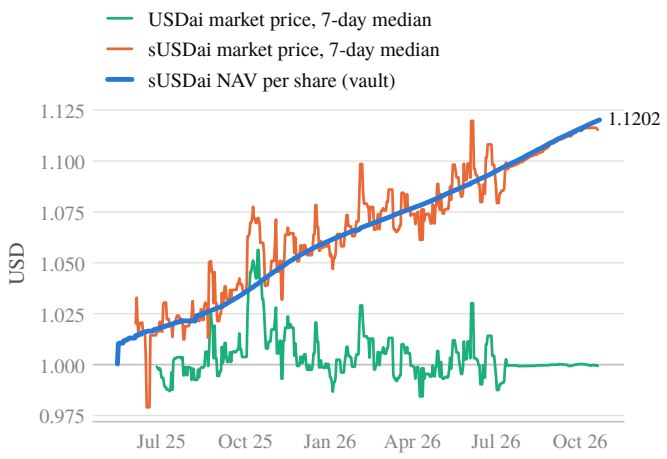


Fig. 5. sUSDai NAV per share from vault deposits and the USDai market price (GoldRush pricing).

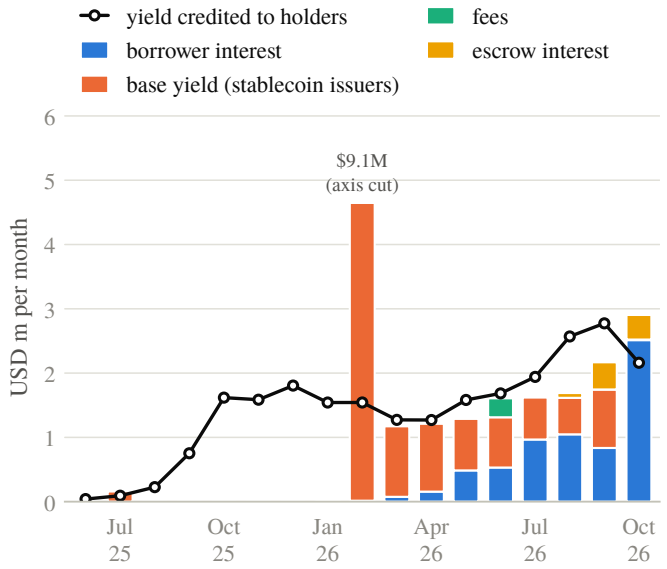


Fig. 6. Yield credited to holders (NAV growth × supply) against cash received by the vault, by month.

and 7.60% over the last 90 days. Multiplying daily NAV changes by the outstanding share supply gives $24.46 M of yield credited to holders.

Against that, the vault recorded $6.65 M of loan interest, $15.08 M of harvested base yield (net of a $1.68 M admin fee), $305 k of origination and repayment fees, and $880 k of escrow interest (net of the router admin fee). The event-based cash-to-NAV-growth ratio is $\kappa = 93.7\%$. Counting only borrower interest and fees, the ratio is 28.4%. Neither measure is a borrower debt-service coverage ratio, and the timing of recognition and receipt can differ. Fig. 6 shows the monthly split. Base yield was the largest recorded historical source, at $15.08 M against $6.65 M of borrower interest. $6.36 M came from PYUSD issuer incentives harvested out of the Base Yield Escrow (a `USDai.Harvested` event in the same transaction). $8.72 M was minted from M^0's wM, the yield-bearing stablecoin that backed USDai early on, including a single $7.82 M deposit on 2026-02-02. The latter one-off event materially affects the full-window comparison.

Table IV separates recurring receipts from two exceptional events. The first is the $7.82 M wM deposit. The second is an October 2, 2026 transaction in which an escrow holding $1,000 for roughly two hours generated approximately $379 k in gross contract-booked "escrow interest", of which approximately $341 k reached the vault after fees. This is not evidence of market-rate interest earned by an ordinary loan. The chain establishes the amounts and their contract classification but cannot determine the transfer's economic purpose. Both exceptional receipts are included in the historical 93.7% coverage ratio and are isolated in the table; excluding them from the numerator is a composition sensitivity, not a forecast of recurring yield. The results establish the historical composition of recognized vault income, not the recurring yield mix after reserve assets and incentive programs change.

Interest also goes to lenders other than the vault. Of the interest paid, $6.65 M (95.8%) went to sUSDai. $294 k went to two other tranche lenders: a junior tranche priced at 28% APR and a one-unit senior position that collects repayment fees. Loans are almost entirely vault-funded (99.95% of current principal).

E. Reconciliation with vault state

At block 517,316,432 the vault reports $555.4 M of assets against 495.7 M shares (Fig. 7). Shares split into 270.8 M on Arbitrum, 204.8 M bridged out and a residual of 20.17 M, which the contract counts as pending redemption. The event scan and state call now use that same block. Our own redemption queue shows 20.17 M pending. Event-derived supply plus that queue differs from `totalShares` by approximately 6×10^{-8} shares, negligible at the displayed precision. This exact-

TABLE IV
LAYERS OF CASH RECEIVED BY THE VAULT RELATIVE TO ACCRUED NAV YIELD. THE FINAL TWO LAYERS ARE EXCEPTIONAL RECEIPTS, NOT ESTABLISHED RECURRING INCOME.

Cash layer (cumulative)	This layer	Cumulative	κ
Borrower interest and fees	\$6.95 M	\$6.95 M	28.4%
Escrow interest on pre-funded loans	\$0.54 M	\$7.49 M	30.6%
Recurring PYUSD issuer incentives (monthly harvests)	\$6.36 M	\$13.85 M	56.6%
Recurring wM reserve yield	\$0.90 M	\$14.75 M	60.3%
One-off wM deposit (2026-02-02)	\$7.82 M	\$22.57 M	92.3%
Protocol transfer booked as escrow interest (2026-10-02)	\$0.34 M	\$22.91 M	93.7%
Credited to holders through NAV		\$24.46 M	100%
Accrued, not yet received	\$1.55 M		6.3%

block identity is a check on share accounting, not independent verification of cross-chain holder balances. The bridged supply is close to the sum of the three EVM spoke snapshots plus Solana (204.6 M). The vault's `pendingLoanBalance` is \$181.8 M. Our stitched loan book exceeds it by \$0.39 M (0.22%). Although the loan accounting includes accrued interest, the available fields do not yet provide an itemized bridge explaining the \$393 k residual. This is a useful consistency check, not proof that every reconstructed loan or off-chain claim is correct.

The assets side breaks down as \$242.3 M of idle deposit balance, \$181.8 M of active loans, \$1.04 M of accrued loan interest, \$0.98 M of unharvested base yield, \$0.66 M of accrued escrow interest and \$128.9 M of `depositTimelockBalance`. That last item is 23.2% of assets, and it is not a loan.

F. Checking GoldRush against the chain

Almost every number in this paper comes from GoldRush, so we pulled the underlying logs again from a public Arbitrum node with `eth_getLogs`, without GoldRush, over the same block range. For each of the 22 event sets the analysis depends on, we compare the two sources log by log, identifying each log by its transaction hash, topics and data, and we compare the sum of the key amount (Table V). 22 of 22 sets are identical, covering 246,262 logs.

The check found one real problem. GoldRush's event endpoint treats `ending-block` as exclusive, so our first scan skipped the last block of every 1,000,000-block window. We confirmed this directly: a log in block B is returned for the range $[B, B{+}1]$ but not for $[B, B]$. We fixed the request, fetched the 182 window-end blocks for every scanned event and re-ran the check. The patch added two Livepeer logs (one ticket payment of 0.0024 ETH) and nothing to the USD.AI events. We also found that, for some transactions, GoldRush numbers logs differently from the node, so we never match or deduplicate logs on their index.

For token supply, `totalSupply` at the snapshot block equals the sum of GoldRush holder balances exactly on Base and is 0.19% higher on Plasma. The public Arbitrum and Ethereum nodes do not serve historical state, so their latest-block comparisons use a different timestamp and differ from the holder snapshots by approximately 0.68% and -0.60%, respectively. These differences should not be conflated with the

TABLE V
GOLDRUSH DATA CHECKED AGAINST A PUBLIC ARBITRUM NODE

Event (Arbitrum)	RPC logs	GoldRush logs	Same
Loans originated, router V1	35	35	yes
Loans originated, router V2	6	6	yes
Loans migrated V1 to V2	17	17	yes
Loans refinanced	3	3	yes
Loans refinanced (3-field)	2	2	yes
Repayments, router V1	118	118	yes
Repayments, router V2	79	79	yes
Lender payouts, router V1	235	235	yes
Lender payouts, router V2	96	96	yes
Fees paid, router V2	61	61	yes
Breaches and liquidations, V2	0	0	yes
Reserve accounts created	25	25	yes
USDai mints	16,377	16,377	yes
USDai redemptions	8,867	8,867	yes
sUSDai deposits	20,093	20,093	yes
sUSDai redemption requests	3,002	3,002	yes
sUSDai redemptions processed	2,841	2,841	yes
Base yield deposited to vault	12	12	yes
Escrow deposits	9	9	yes
Escrow withdrawals	4	4	yes
Escrow cancellations	3	3	yes
Livepeer winning-ticket payments	194,377	194,377	yes
sUSDai supply: `totalSupply` vs. sum of GoldRush holders			
Arbitrum (latest block)	268.98 M	270.82 M	+0.68%
Ethereum (latest block)	102.14 M	101.53 M	-0.60%
Base (same block)	3.64 M	3.64 M	+0.00%
Plasma (same block)	26.16 M	26.11 M	-0.19%

negligible same-block share-accounting residual above. The escrow redemption receipt, re-read from the node, shows the same burn of 128,895,244.1 USDai and the same PYUSD payment to the same wallet. At the later block 518,591,558, the USDai contract held \$266.3 M of stablecoins (\$266.3 M of it PYUSD) against \$266.3 M of USDai counted across the hub and EVM spokes, a measured coverage of 100.00% for that scope and instant. This observation does not by itself prove redemption availability under stress, recovery of sUSDai credit claims, or an omnichain reserve audit.

G. Escrow pre-funding

The vault made 9 escrow timelock deposits totaling \$223.1 M. Most closed normally: 7 were either withdrawn into an originated loan or cancelled (3 cancellations, which returned principal plus interest to the vault). 2, both made on 2026-09-23 and worth \$128.9 M, carry a `loanTermsHash` that no Loan Router has originated and that has not been cancelled. Together they equal `depositTimelockBalance` to the cent. They earn an escrow rate of 6.5%, which the vault accrues optimistically. None of the timelock contracts holds any tokens: `balanceOf` returns zero for USDai, PYUSD and USDC.

Fig. 8 and Table VI show the traced transfers. Every escrow deposit moves USDai from the vault to a single address we call the escrow facility (\$223.1 M in total, of which cancellations returned part). It is a Gnosis Safe, and its transactions are submitted by accounts also observed submitting the protocol's multisig transactions. This suggests operational overlap, not

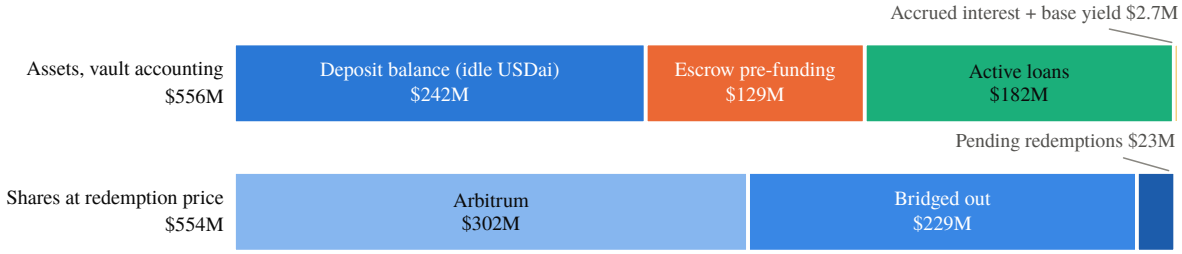


Fig. 7. The vault's balance sheet at block 517,316,432, read with `eth_call`. Shares are valued at the redemption share price.

TABLE VI
LARGEST STABLECOIN FLOWS AROUND THE ESCROW FACILITY AND THE PYUSD HUB. HUB OUTFLOWS ALSO INCLUDE PYUSD THE HUB RECEIVED FROM THE LARGEST BORROWER AND THE MULTISIG SUBMITTER.

From → To	Asset	USD
sUSDai vault (`0x0b2b...5ef9`) → escrow facility	USDai	$223.1 M
PYUSD hub (`0x1168...21b1`) → off-ramp A (`0xf845...abb7`)	PYUSD	$339.6 M
escrow facility (`0x844b...6286`) → PYUSD hub (`0x1168...21b1`)	PYUSD (redeemed)	$142.4 M
escrow facility (`0x844b...6286`) → PYUSD hub (`0x1168...21b1`)	PYUSD	$62.2 M
escrow facility (`0x844b...6286`) → sUSDai vault (`0x0b2b...5ef9`)	USDai	$16.3 M
escrow facility (`0x844b...6286`) → `0x24a9...1392`	PYUSD	$6.0 M
escrow facility (`0x844b...6286`) → `0xfd1c...ca91`	USDai	$5.9 M
`0xfd1c...ca91` → PYUSD hub (`0x1168...21b1`)	PYUSD (redeemed)	$5.9 M
PYUSD hub (`0x1168...21b1`) → service wallet B (`0x264b...97b5`)	PYUSD	$5.5 M
escrow facility (`0x844b...6286`) → `0x6034...23ef`	USDai	$5.1 M

verified common legal ownership. For the two open deposits, the facility redeemed USDai into PYUSD at the USDai contract on the same day and paid PYUSD to a wallet we call the PYUSD hub. The hub was first active on 2026-04-29 and has 56 transactions. It also received PYUSD from the largest borrower and from an account that submits the protocol's multisig transactions. It transferred large amounts to a high-activity destination labeled "off-ramp A" in the figure, which holds PYUSD and USDG and is active on 19 chains, and smaller amounts to another service-like wallet. Neither label identifies a verified custodian or ultimate beneficiary. The fee recipient also routes PYUSD to the high-activity destination.

Escrow pre-funding is contemplated by the contracts and may correspond to a signed facility not yet originated on-chain. At the observed snapshot, $128.9 M of reported sUSDai assets represents two accounting claims with no corresponding originated loan or wrapped GPU collateral in the analyzed contracts. The tokens were redeemed and transferred through external wallets. Because PYUSD is fungible, following gross transfers through a commingled wallet does not establish ownership or the destination of each deposited dollar. The chain alone cannot determine whether the vault has segregated funds, contractual repayment rights, or other off-chain security. Optimistic escrow yield is accrued before a matching on-chain loan exists; legal status, recoverability and subsequent settlement require independent verification and a response from the protocol (Section V).

TABLE VII
SUSDAI HOLDER CONCENTRATION BY CHAIN

Chain	Holders	Supply	Top-1	Top-10	HHI
Arbitrum	3,778	270.8 M	33.3%	86.7%	0.209
Base	468	3.6 M	95.1%	99.6%	0.905
Ethereum	135	101.5 M	81.6%	100.0%	0.699
Plasma	1,424	26.1 M	97.6%	99.9%	0.953

H. RQ3 and RQ6: Holders and the cross-chain footprint

On Arbitrum, sUSDai has 3,778 holder addresses. The largest address holds 33.3% and the top ten hold 86.7% (address-level HHI 0.209, Gini 0.994). Table VII gives the same statistics for each chain, and Table VIII lists the largest hub addresses with contract labels from GoldRush. On every chain, a handful of contracts and pools custody much of the supply (Fig. 9). Address concentration is not beneficial-owner concentration: bridges, pools and vaults may represent many end users. The largest identified DeFi venue is Pendle, which holds 13.2% of Arbitrum sUSDai. On Plasma, 44.1% of USDai sits in Pendle and another 20.0% in vault strategies.

Supply is spread across five chains: 270.8 M on Arbitrum, 101.5 M on Ethereum, 26.1 M on Plasma, 3.64 M on Base and 73.3 M on Solana (Fig. 10). Only one address holds at least 100,000 sUSDai on more than one EVM chain. It is a high-activity contract with more than 8 million Arbitrum transactions, and it is the largest or second-largest holder on all four EVM chains, with $>95\%$ of supply on Base and Plasma. Historical snapshots at monthly blocks show when the Ethereum footprint appeared. sUSDai on Ethereum was a test-sized position (2 holders in 2025-07-01, under 0.1 M shares on 2026-06-01) until mid-2026. It then grew to 89.3 M shares and 102 holders by 2026-10-01 (Fig. 11). Two addresses hold 99.7% of it: the cross-chain contract above and the Morpho Blue singleton (18.0%), where sUSDai is posted as loan collateral. GoldRush's cross-chain activity endpoint shows where borrowers and top holders operate (Fig. 12). The 10 borrowers with activity data have used between 4 and 12 chains (median 7), so borrowers are active well beyond Arbitrum. The 66 top holders range from 1 to 29 chains (median 10). Many single-chain holders are pools and vault contracts.

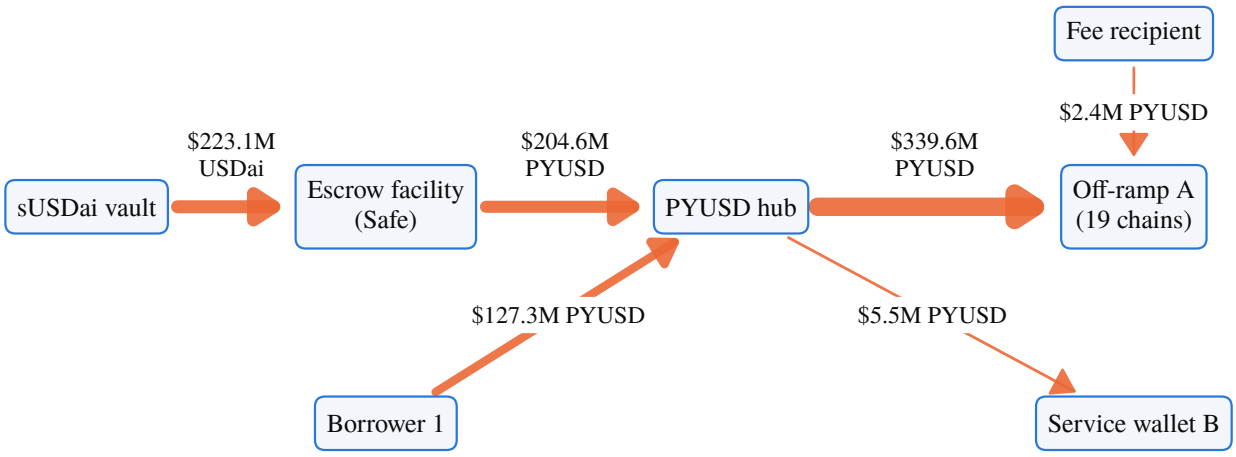


Fig. 8. Where escrowed vault funds went. Edge width is proportional to USD. All flows are decoded ERC-20 transfers or USDai redemptions on Arbitrum. Wallets are labeled by on-chain role only.

TABLE VIII
LARGEST SUSDAI HOLDERS ON ARBITRUM

Holder	sUSDai	Share	Label
0x7ced...20af	90.16 M	33.3%	unlabeled
0x52aa...e497	75.43 M	27.9%	unlabeled
0x30cc...2f62	35.68 M	13.2%	SY Staked USDai
0xed41...6029	8.89 M	3.3%	unlabeled
0xe7f5...e8e2	6.60 M	2.4%	unlabeled
0x9167...6700	5.38 M	2.0%	unlabeled
0xbeed...6dba	4.56 M	1.7%	unlabeled
0xa520...a575	2.82 M	1.0%	unlabeled

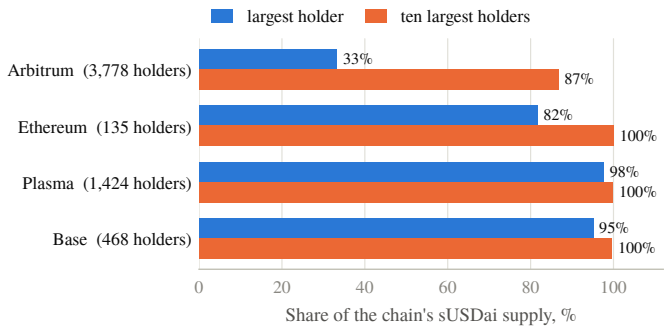


Fig. 9. Share of each chain's sUSDai held by its largest holder and its ten largest holders.

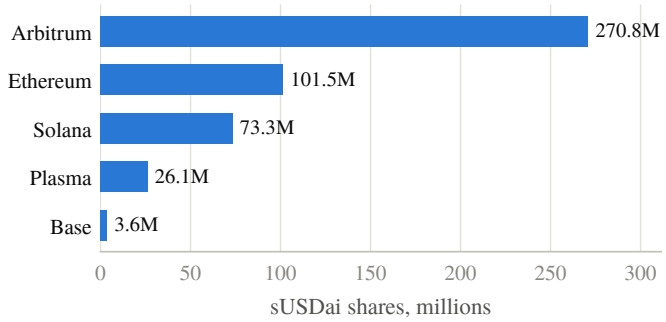


Fig. 10. sUSDai shares by chain (holder snapshots; Solana from mint supply).

TABLE IX
BORROWERS (ADDRESS PREFIXES), SORTED BY ORIGINATED PRINCIPAL

Borrower	Loans	Originated	Outstanding	Interest paid
0x6444...d39d	3	$132.40 M	$41.71 M	$2,995 k
0x7f12...8865	1	$54.40 M	$54.40 M	$356 k
0xefc6...e332	3	$36.01 M	$32.42 M	$1,373 k
0xaae6...25b0	1	$26.82 M	$22.35 M	$1,123 k
0xc991...5398	3	$18.93 M	$11.70 M	$585 k
0x9243...1960	1	$9.56 M	$9.56 M	$58 k
0xd08f...359a	1	$7.50 M	$7.32 M	$121 k
0x8db6...2fb9	1	$2.30 M	$1.79 M	$206 k
0xe277...0eb9	7	$0.54 M	$0.43 M	$39 k
0x6726...be31	1	$0.53 M	$0.40 M	$31 k
0x7c5e...854e	1	$0.48 M	$0.00 M	$23 k
0x7b79...4462	15	$0.44 M	$0.00 M	$15 k
0x45b3...94d3	1	$0.20 M	$0.16 M	$15 k

I. RQ4: Borrower concentration

The loan book is concentrated. The largest borrower address accounts for 45.6% of originations and the top three for 76.8%. HHI is 0.273 on originations and 0.197 on outstanding principal. The latter is more relevant to current loss exposure; neither statistic establishes independence of the underlying legal borrowers. HHI is used here as a portfolio concentration metric, not an antitrust-market classification. Table IX lists each borrower. Rates run from 7.0% to 15.0%, with an outstanding-weighted average of 10.26%. 25 reserve accounts map back to borrowers through the factory. One reported $26.82 M facility matches, to the reported precision, a single loan of $26,821,644 originated on 2026-04-06. We do not attribute any address to an off-chain entity in our results.

J. RQ5: Redemptions and GPU prices

Holders filed 3,002 redemption requests from 2,545 addresses, for 350.3 M shares. Among completed requests, the

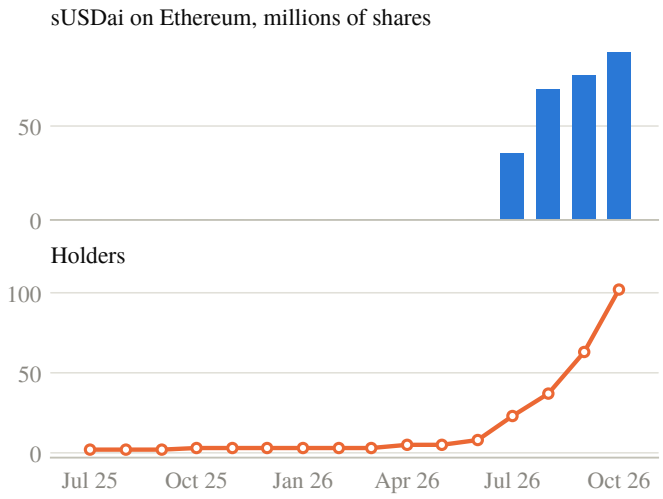


Fig. 11. sUSDai on Ethereum over time, from historical holder snapshots at the first block of each month.

median wait was 13.0 days, the 90th percentile 26.6 days, and the longest 127.3 days. Another 20.17 M shares remained pending at the cutoff; their final wait times are right-censored. These completed-request statistics do not measure the unconditional exit-time distribution or dollar-weighted experience. Servicing is lumpy (Fig. 13), with large batches: three calls on 2026-03-07 and 2026-04-06 together paid out about $217 M. Priced in PYUSD, which reduces feed noise common to both tokens, USDai never traded below 0.975 (on 2025-08-19). Selling on a DEX provides another route to exit, subject to market depth and price.

The available weekly series does not identify a causal effect of GPU prices on redemptions. Over 53 overlapping weeks, Spearman's ρ between weekly requested shares and the H100 rental price is -0.41 in levels and 0.01 in weekly changes (Fig. 14). Protocol growth and declining rental-price trends confound the level correlation. The near-zero change correlation is descriptive only; it neither establishes independence nor rules out lagged responses, benchmark mismatch or effects conditional on contractual revenue exposure.

K. RQ7: GPU credit and observable compute payments

Over the same window, Livepeer gateways paid orchestrators 371 ETH ($1.01 M) in 194,377 winning tickets. 21 gateways paid 104 orchestrators. The largest gateway accounts for 49.4% of payments and the top three for 83.1% (HHI 0.318). On the supply side, payments are spread out (HHI 0.050). Livepeer's payments mix video transcoding and AI inference, and the events do not separate the two.

Akash, measured through its own API, recorded $3.47 M of spend in the window and $6.5 M all-time. Active leased GPUs ranged from 40 to 529. On 2026-10-22, the network reported 263 of 390 GPUs active across 58 providers (Fig. 17). The median Akash H100 price was $2.56/hr and the A100 $1.49/hr, against a centralized-cloud H100 median of $3.36/hr (Fig. 18).

TABLE X
GPU CREDIT AND DECENTRALIZED COMPUTE, STUDY WINDOW

Metric (study window)	USD.AI	Akash	Livepeer
USD volume	$290.1 M lent	$3.47 M spent	$1.01 M paid
Revenue to capital/providers	$6.94 M interest	$3.47 M	$1.01 M
Demand-side counterparties	13 borrowers	n/a	21 gateways
Supply-side counterparties	4 tranche lenders	58 providers	104 orchestrators
Top-1 share	45.6%	n/a	49.4%
Data source	GoldRush (Arbitrum)	Akash Console API	GoldRush (Arbitrum)

Fig. 16 and Table X put the three observed flows side by side on a log scale. USD.AI's gross loan originations are 65 times the $4.47 M of measured Akash and Livepeer payments over the window. This ratio compares capital advanced with operating payments, and the borrower populations need not overlap. It cannot be interpreted as a ratio of credit to underlying compute demand, nor does the sample span all decentralized or private compute transactions. USD.AI borrower-address concentration (HHI 0.273) and Livepeer gateway concentration (HHI 0.318) describe different participant populations (Fig. 19). The compute use financed by the reconstructed loans is not independently linked to on-chain rental or inference receipts in this dataset.

L. What the debt implies about the compute

The chain records principal, contractual rates and maturities, but not actual GPU rental revenue. The current outstanding balance implies approximately $18.7 M of annual interest at the stated rates. If, as an illustrative assumption rather than a recovered contractual payment schedule, principal amortizes straight-line to maturity, annual principal payments would be $71.2 M and total modeled debt service $89.9 M. Our earlier work treated machine labor as hours of work rather than units of hardware [12], [18]; that frame permits a transparent sensitivity calculation. At the Akash H100 median of $2.56 per hour, $89.9 M would require 35.1 M billable GPU-hours per year before operating expenses. At 70% utilization, this is the equivalent of 5,729 continuously available H100s and 8.6 MW at an assumed 1.5 kW per GPU before cooling. At the centralized-cloud H100 price of $3.36, the equivalent is 4,365 GPUs (6.5 MW); at a hypothetical $1.50 per hour, it is 9,778 (14.7 MW). Fig. 15 plots this H100-equivalent sensitivity, not the actual required B200/B300 deployment. The pledged hardware is model-heterogeneous and its realized prices, utilization, customer terms, power cost and scheduled principal payments are not observed. Dividing the modeled debt service by 4,560 GPUs at 70% utilization yields $3.22 per GPU-hour ($2.25 at full utilization), before any operating expense. This is a conditional gross-revenue hurdle, not evidence that the fleet can service debt.

The hypothetical threshold can be tested only with independently verified unit counts, contractual repayment schedules, utilization, realized receipts, operating costs, customer commitments and collateral rights. Publication of a hardware-token record alone is insufficient for that underwriting test.

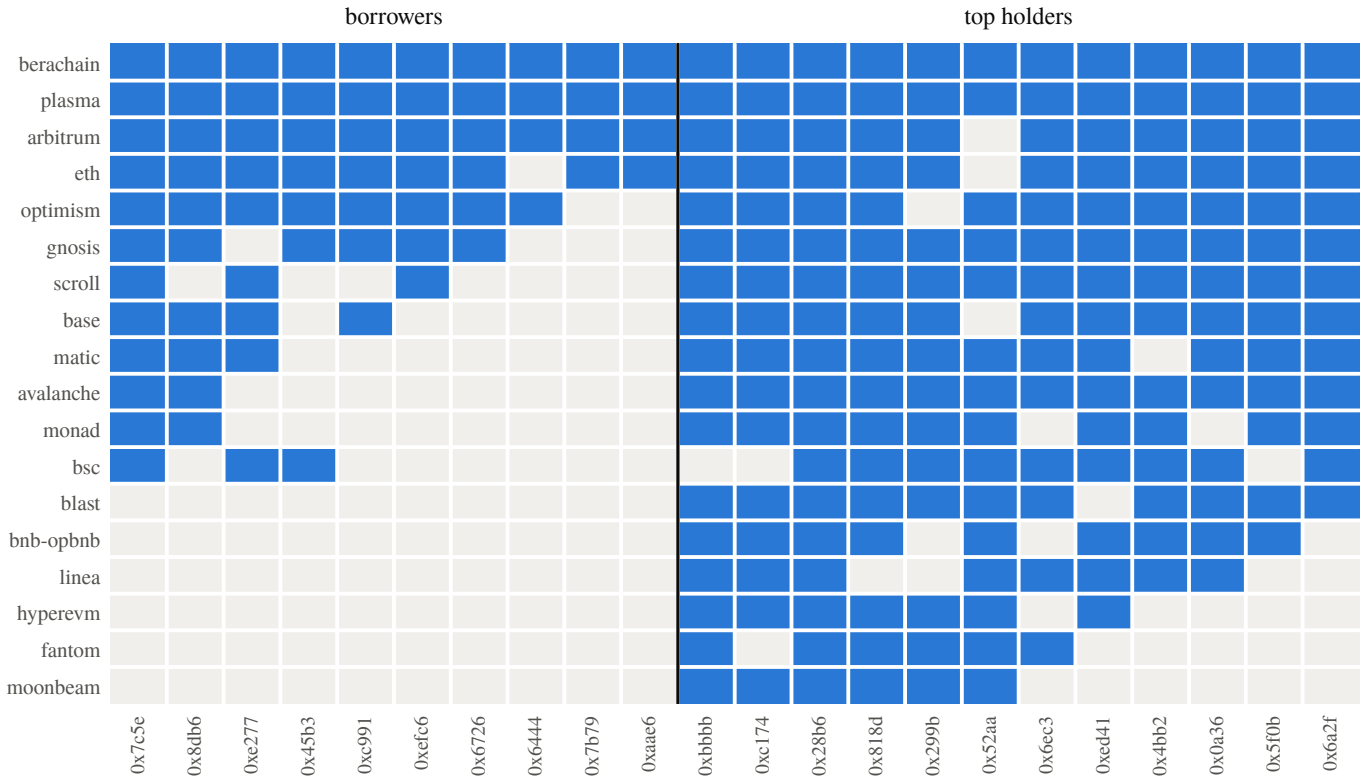


Fig. 12. Cross-chain activity of borrowers and top holders from GoldRush's address activity endpoint. Each filled cell is a chain on which the address has transacted.

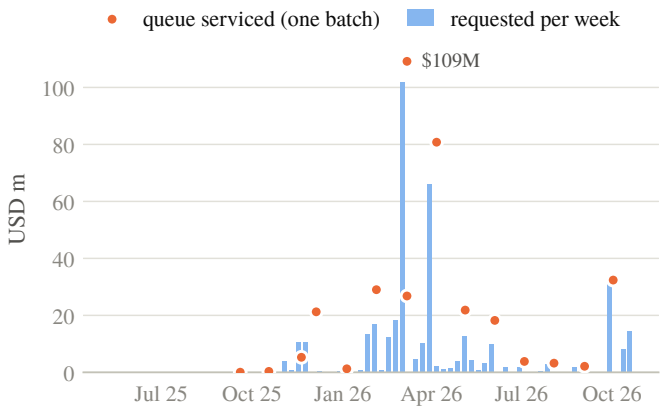


Fig. 13. Weekly redemption requests (USD at NAV) and queue servicing batches.

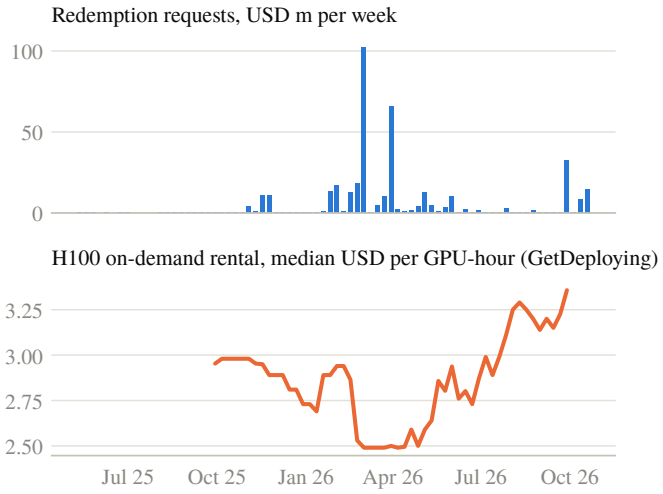


Fig. 14. Weekly redemption requests against the median H100 on-demand rental price (GetDeploying).

V. Discussion

A. What the chain can and cannot tell a lender

Chain data is strong on recorded liabilities and token movements. It permits a reconstructed loan book, borrower-address ledger, repayment history and redemption queue, with an exact-block comparison to selected vault accounting views. It is weak on physical assets and contractual rights. It cannot establish that a GPU exists, that it is generating revenue, or that a lender can repossess it. Three findings sit at that boundary. First, borrower interest and fees contributed 28.4% of NAV-accrued yield over this window. Stablecoin incentives and two extraordinary transfers account for much of the remainder, so the historical 93.7% cash-coverage ratio cannot be interpreted as recurring borrower cash generation. Second, 23.2% of reported assets was attributed to pre-funded escrow without a matching on-chain loan or wrapped collateral at

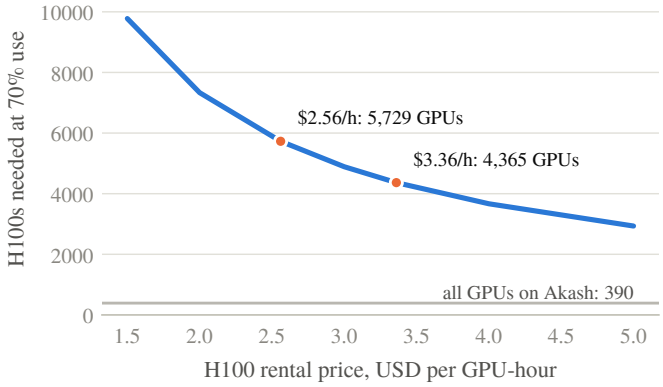


Fig. 15. Illustrative H100-equivalent GPUs required at 70% utilization to cover modeled straight-line debt service, before operating costs. The Akash network's GPU count is shown for scale, not as a comparable borrower fleet.

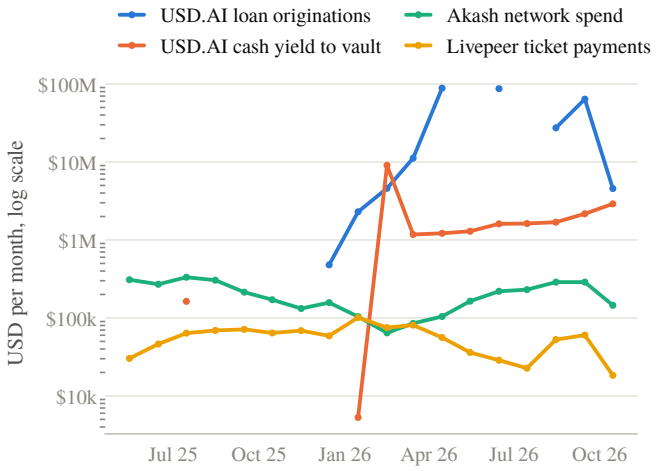


Fig. 16. Observed monthly USD flows on a log scale. Loan originations, lender receipts and compute payments are different economic quantities and are not estimates of the same market.

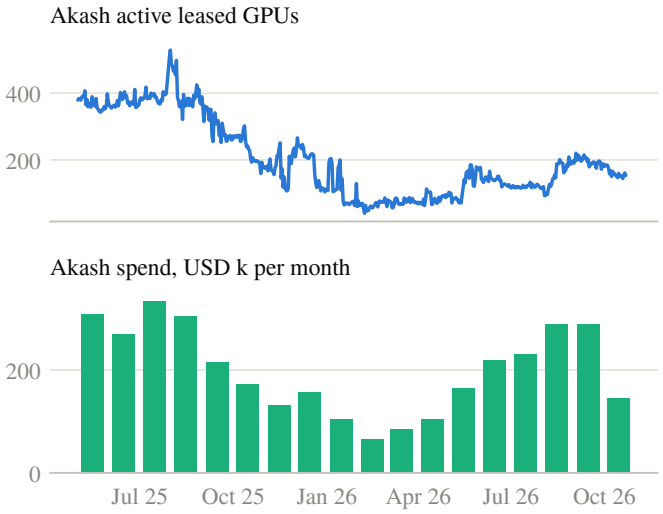


Fig. 17. Akash active leased GPUs and monthly spend (Akash Console API).

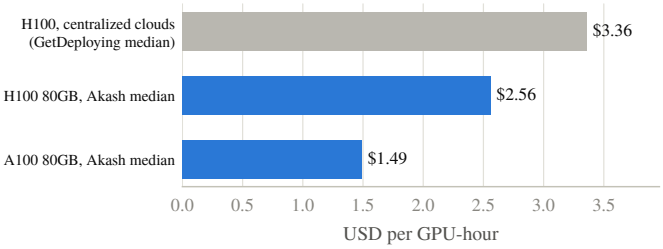


Fig. 18. GPU rental prices: decentralized (Akash median) and centralized (GetDeploying median), USD per GPU-hour.

the snapshot; the off-chain contractual claim is unknown. Third, loan maturities extend 2.6 more years on average while holders can request redemption through a periodically serviced queue. The snapshot idle balance provides near-term liquidity, but a severe run would require further funding, scheduled repayments or asset sales. The headline NAV alone does not resolve these risks.

B. What a neocloud unwind would do

The borrowers are GPU operators, some of which participate in the broader neocloud sector. Published commentary estimates more than $20 B of GPU-backed debt across that sector [19], but this is context, not a verified denominator for the present loan book. H100 posted rental prices declined substantially from early 2024 to late 2025 in the cited sources [19], and the GetDeploying sample used here has a centralized median of $3.36. Posted H100 rates do not reveal the realized contracted revenue of B200- and B300-backed borrowers. Hardware-life assumptions also vary with workload, resale markets and maintenance [12].

USD.AI provides an observable subset of GPU-backed credit. Table XI applies hypothetical recoveries to outstanding loan claims at block 517,316,432, not to the independently appraised resale value of collateral. It is arithmetic, not a credit model. It omits recovery delays, enforcement costs and correlated equipment-price declines. The protocol's documentation also describes possible residual-value insurance, but this study has not verified any policy's coverage, exclusions, claimant rights or insurer solvency [13]; the table therefore assumes no incremental insurance recovery. The largest borrower address holds 29.9% of outstanding principal and the three largest 70.5%. At a 40% recovery of the debt claim, a default by the largest costs 5.9% of assets and leaves modeled NAV at 1.054; the three largest cost 13.9% and leave NAV at 0.965. If every borrower defaults at the same assumed recovery, NAV falls to 0.900. If all open escrow value is separately written off, modeled NAV is 0.640; that is a deliberately severe combined scenario, not an estimate of its probability.

Three features condition how a loss might propagate. First, the vault prices deposits using optimistic NAV and redemptions using conservative NAV. The documented default and recovery process can involve delayed recognition of actual

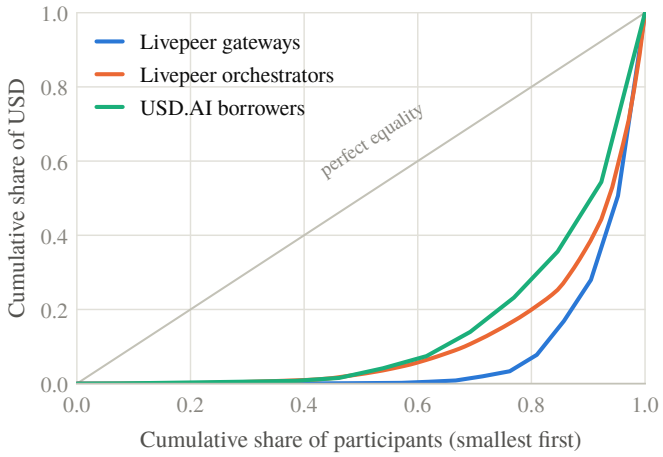


Fig. 19. Concentration of payers and payees: Livepeer gateways and orchestrators against USD.AI borrowers.

TABLE XI
ILLUSTRATIVE VAULT LOSS SCENARIOS. RECOVERY PERCENTAGES APPLY TO OUTSTANDING LOAN CLAIMS, NOT TO GROSS STATED GPU COLLATERAL VALUES; INSURANCE PROCEEDS ARE EXCLUDED.

Defaults	Recovery	Loss	Share of assets	NAV after
largest borrower	60%	$21.8 M	3.9%	1.076
three largest borrowers	60%	$51.4 M	9.3%	1.017
every borrower	60%	$72.9 M	13.1%	0.973
largest borrower	40%	$32.6 M	5.9%	1.054
three largest borrowers	40%	$77.1 M	13.9%	0.965
every borrower	40%	$109.3 M	19.7%	0.900
largest borrower	20%	$43.5 M	7.8%	1.033
three largest borrowers	20%	$102.8 M	18.5%	0.913
every borrower	20%	$145.8 M	26.3%	0.826
every borrower; escrow not returned	40%	$238.2 M	42.9%	0.640

proceeds, so the sequence between missed payments, formal default, recovery and redemptions matters [13]. Second, exits are queued. At the snapshot, idle deposit balance covers the pending queue 11 times over, but this is a point-in-time ratio, not a stressed liquidity guarantee. Third, sUSDai is held in other DeFi contracts: 18.0% of Ethereum supply is at Morpho Blue and 13.2% of Arbitrum supply at Pendle. Depending on collateral parameters, leverage and market depth, a NAV impairment could induce liquidations or discounts; the present data does not establish their magnitude.

Related GPU-financing structures may share sensitivity to equipment resale prices and customer demand [20], although exposure varies by borrower contracts, collateral, security rights and funding terms. Three on-chain questions remain useful: how much realized vault income comes from borrower payments rather than reserve yield; how concentrated the current loan exposures are; and how queued redemptions compare with liquid assets. These indicators can be monitored from the reconstructed ledger, while their interpretation requires off-chain contract and operating evidence.

C. Risks the chain shows only in outline

Seven risks are visible on-chain but cannot be sized from it.

Related parties. Some submitting accounts recur across the protocol multisig and escrow-facility transactions. The largest borrower address, fee recipient and escrow-related wallets also transact with a common high-activity destination. This supports a question about control and economic affiliation, not a conclusion of common ownership or impropriety. A lender cannot treat 13 borrower addresses as 13 legally independent credit exposures without an entity map.

Key compromise. Administrator permissions and Safe signing arrangements create operational dependencies. Their actual security depends on signer custody, quorum, delay mechanisms and incident response, none of which is established by counting on-chain transactions. We have not tested the operators' private-key controls.

Fixed loans, floating benchmark. The loans are fixed at 7.0% to 15.0% for 2.6 more years. The yield holders compare against moves: Treasury bills, and the PYUSD and M^0 incentives that supplied most of the cash yield so far. If those fall, the vault's spread widens; if they rise, the vault holds below-market paper and the queue fills.

Custody, title and valuation. The hardware is described through token metadata down to server models and, for some early loans, serial numbers (Section IV-B). What is not independently established is the unit's existence, location, perfected security interest, liquidation priority, insurance coverage, or whether stated values are appraisals or purchase prices. Book LTV of 66.8% is therefore a ratio of contractual principal to protocol-stated collateral value, not a verified liquidation LTV.

Bridged exits. 41.3% of shares live on Ethereum, Base, Plasma and Solana. A holder there must bridge back to Arbitrum before joining the queue. A paused bridge pauses the exit for two fifths of the supply.

Leverage on top. Pendle holds 13.2% of Arbitrum sUSDai and Morpho Blue 18.0% of Ethereum sUSDai. Fixed-yield buyers in Pendle have locked in today's rate; a NAV mark hits them first, and the leveraged positions sit there.

Classification. This is pooled private credit with a yield-bearing token. Payment-stablecoin rules in the United States now distinguish tokens that pay yield from those that do not, and the treatment of a structure like this one is not settled. A ruling could change who may hold it or how it may be marketed.

D. What this means for different readers

Traders. Three observable indicators deserve monitoring: delays in scheduled repayments (currently the longest observed gap is 25 days), the queued-redemption to idle-cash ratio (currently 11× covered), and sUSDai's market discount to its accounting NAV. The repository computes these indicators; their ability to predict losses or market discounts has not yet been evaluated out of sample. The conservative redemption price (1.1172) and deposit price (1.1203) also differ because of the vault's valuation conventions.

Institutional allocators. Review the two open escrows, which represent 23.2% of reported assets but have no matching

originated loan at the snapshot. Verify legal claims, beneficiaries, expiry provisions and any collateral or insurance independently. The borrower-interest-and-fee ratio (28.4%) is a historical contribution to NAV growth, not a stand-alone sustainable APY or debt-service coverage measure. Underwrite the 29.9% single-name share of current principal, reconcile protocol-stated 66.8% with appraised liquidation values, and assess insured and uninsured recovery paths separately (Table XI).

Neoclouds. Public event logs can lower the cost of verifying payment and ownership records, although this study does not compare audit costs with bank credit. Borrowers could make the asset side easier to underwrite by providing independently attestable utilization, contractual revenues, power expense, location, title and recovery rights. Whether such disclosure would reduce borrowing costs relative to the observed 10.26% weighted-average coupon remains an empirical question.

Researchers. The included frozen extracts match their SHA-256 manifest and support the reported reconciliations. The small data handover does not contain the complete Livepeer extract or full analysis pipeline, so full offline reproduction additionally requires those materials. The method may transfer to other GPU-credit protocols with equivalent event and state coverage. Open questions include independent collateral valuations, borrower revenue, and recoveries once a loan defaults. A second protocol and a longer window would make this a comparative study.

E. Monitoring indicators

The pipeline can run on a schedule. We suggest six indicators, each computed by code in the repository:

1) *Cash coverage* κ by month, decomposed into borrower receipts, recurring reserve income, and exceptional transfers.
2) *Pre-funded share*: `depositTimelockBalance` over `totalAssets`, with the age of each open escrow.
3) *Borrower HHI* on outstanding principal, and the largest borrower's share.
4) *Queue pressure*: pending redemptions against idle deposit balance, and the age of the oldest pending request.
5) *Missed payments*: loans whose last repayment is older than their repayment interval.
6) *Peg stress*: the market discount of USDai and sUSDai to NAV.

F. Other GPU-backed protocols

Other protocols also tokenize GPU financing or GPU revenue, and some issue their own synthetic dollars, such as GAIB's AID [20]. The same method works for any of them whose contracts run on a chain GoldRush indexes: decode from source, rebuild the loan book, reconcile against state and trace the cash. Comparing them on cash coverage, pre-funding and borrower concentration is a natural next study.

G. Data coverage

Without a unified API, several steps would have needed an archive node and custom indexing. GoldRush provided complete decoded histories for the lending contracts, event scans across 182 block windows, holder lists on four chains, holder lists at past blocks, one call listing every chain an address has used, cross-chain wallet balances and daily token prices. Three gaps remain. GoldRush does not offer `eth_call`, so the state check used a public RPC. Holder lists at past blocks are available only on its Foundational chains, so we have them for Ethereum but not Arbitrum. Akash and other Cosmos chains are not indexed.

H. Limitations

We study one protocol over 17 months. The loan book is reconstructed from selected events, and both API and node checks use those same event definitions. Identical raw logs do not independently prove complete contract coverage, correct borrower identities or the economic meaning of every transaction. The event cutoff and primary state check now share a block, but the \$0.39 M loan-balance residual remains unexplained by an itemized accounting bridge. The reserve observation uses a later block, and the Arbitrum and Ethereum holder-to-supply checks have unequal observation times. Two V1 topic hashes remain unidentified. Hardware-token metadata is mutable and was retrieved after the study cutoff; two host-served records are not minted. The aggregate GPU count includes mixed-unit rows requiring further inspection. Holder-address concentration is not beneficial-owner concentration. Completed redemption waits exclude still-pending requests, and most loans have not matured. The H100 price series is neither model-matched to the B200/B300-heavy collateral nor a record of borrower realized revenue. Livepeer payments mix transcoding and AI inference; Akash figures come from its own API. The current small handover archive omits the large Livepeer raw extract, which must be supplied separately for a fully offline reconstruction. We have not independently verified executed loan documents, title, collateral custody, enforceable insurance, leases, utilization, operating expenses or customer receivables. No off-chain person or entity is attributed to a wallet; the data do not demonstrate misuse of funds.

I. Ethics and right of reply

All data is public. We do not name individuals or link wallets to companies. We invite corrections and an explanation of the escrow accounting from USD.AI and will incorporate any substantiated response in a revised version.

VI. Conclusion

Event logs allow a detailed reconstruction of recorded GPU-backed lending and vault flows: \$290.1 M in gross loan originations to 13 borrower addresses, historical income from borrower payments and base yield, lumpy redemption servicing, and 23.2% of reported assets attributed to pre-funded escrows without matching on-chain loan originations

at the snapshot. These are financial-observability findings, not evidence that credit has outrun underlying compute demand. Their significance is the separation between auditable financial records and off-chain collateral, contracted revenue and legal recovery. A credible machine-credit risk measure will require both sets of information. Comparing this framework across protocols and testing whether its indicators forecast repayment, liquidity or recovery outcomes remain tasks for further research.

Disclosures

Conflicts of interest: The author has no financial relationship with USD.AI, GAIB, or any protocol, borrower or lender discussed in this paper, and holds no position in their tokens. **Use of AI:** AI assistants (Claude and ChatGPT) assisted with code, text drafting, and editorial consistency checks. The author is responsible for the claims, methods and interpretation. **Data:** GoldRush API by Covalent; Akash Console API; GetDeploying GPU price snapshot (CC BY 4.0) [17]; Arbitrum and Solana public RPCs.

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